

8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (XR8.6a)$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- We are curious whether holding education, experience, and $METRO$ constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i | \mathbf{x}_i, FEMALE = 0) = \sigma_M^2$ and $\text{var}(e_i | \mathbf{x}_i, FEMALE = 1) = \sigma_F^2$. We specifically wish to test the null hypothesis $\sigma_M^2 = \sigma_F^2$ against $\sigma_M^2 \neq \sigma_F^2$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.
- We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i | \mathbf{x}_i, MARRIED = 0) = \sigma_{SINGLE}^2$ and $\text{var}(e_i | \mathbf{x}_i, MARRIED = 1) = \sigma_{MARRIED}^2$. Specify the null hypothesis $\sigma_{SINGLE}^2 = \sigma_{MARRIED}^2$ versus the alternative hypothesis $\sigma_{MARRIED}^2 > \sigma_{SINGLE}^2$. We add $FEMALE$ to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

- Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.
- Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

- The OLS fitted model from part (b), with usual and robust standard errors, is

$\widehat{WAGE}$	$= -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$
(se)	(2.36) (0.14) (0.031) (1.05) (0.81)
(robse)	(2.50) (0.16) (0.029) (0.84) (0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- If we add $MARRIED$ to the model in part (b), we find that its t -value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

8.6

(a)

$$1. H_0: \sigma_M^2 = \sigma_F^2 \quad \text{v.s.} \quad H_1: \sigma_M^2 \neq \sigma_F^2$$

$$2. F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \sim F_{577-4, 423-4}$$

$$\because \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} > 1 \quad \therefore \text{比较右尾}$$

$$3. \alpha = 0.05$$

$$RR = \{F > F_{0.025}(573, 419) = 1.1968\}$$

$$4. F^* = \frac{169.567}{12.0242} = 1.1729$$

$$\hat{\sigma}_M^2 = \frac{SSE}{N-K} = \frac{97161.9174}{573} = 169.567$$

5. Because $F^* = 1.1729$ doesn't fall into reject region, we fail to reject H_0 . There isn't evidence to support H_1 is true.

(b)

$$1. H_0: \sigma_{\text{single}}^2 = \sigma_{\text{married}}^2 \quad \text{v.s.} \quad H_1: \sigma_{\text{married}}^2 > \sigma_{\text{single}}^2$$

$$\sigma_{\text{single}}^2 = \frac{56231.0382}{400-5} = 142.3571$$

$$\sigma_{\text{married}}^2 = \frac{100703.0471}{600-5} = 169.2488$$

$$2. F = \frac{\sigma_{\text{married}}^2}{\sigma_{\text{single}}^2} \sim F_{595, 395}$$

$$3. \alpha = 0.05$$

$$RR = \{F > F_{0.05}(595, 395) = 1.1647\}$$

$$4. F^* = \frac{169.2488}{142.3571} = 1.1889$$

5. Because $F^* = 1.1889$ falls into reject region, we reject H_0 is true.
There is evidence to support H_1 is true.

(4) ① $H_0: \alpha_2 = \dots = \alpha_5 = 0$ v.s H_1 : Not all the α_i in H_0 are zero

$$X^2 = N \times R^2 \sim \chi^2_{(4)}$$

$$\alpha = 0.05 \quad RR = \{X^2 > \chi^2_{0.95}(4) = 9.488\}$$

$$\because NR^2 = 59.03 > RR = 9.488$$

$\therefore$ We reject H_0 is true. There is evidence to support H_1 is true.
Thus, model exists heteroskedasticity.

② 否. 因為 married 沒有包含在 NR^2 test 的變數中. 無法證明
誤差變異數是否因為婚姻狀態而不同.

(d) White test 的輔助迴歸將殘差平方

原解釋變數 (4 個). 原解釋變數的平方項 (2 個, 扣除虛設變數)
所有解釋變數的交叉項 (6 個)

$$\Rightarrow df = 4 + 2 + 6 = 12$$

$$\chi^2_{0.95}(12) = 21.026$$

$\because 78.82 > 21.026$. 落入拒絕域. 拒絕 H_0 . 模型存在異變異數

(e) 區間變窄 $\rightarrow$ 標準誤差小: EXPER, METRO, FEMALE

區間變寬 $\rightarrow$ 標準誤差大: Intercept, EDUC

否, 因為異變異效對於不同係數的 OLS 估計值的抽樣變異的影響方向並不相同。

存在異變異效時, 應該以 robust standard errors 作為推論。

(f) 此結果和 (b) 沒有衝突。

因為 (b) 是在檢定誤差變異效, 結果顯示已婚和單身的誤差變異效不同, (f) 是在檢定條件平均效, 其結果不顯著, 表示 MARRIED 對於預測工資沒有顯著的解釋能力。

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.
- Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?
- Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.
- Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?
- Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

1a)
$$\hat{MILES} = -391.548 + 14.201 \times INCOME + 15.741 \times AGE - 81.826 \times KIDS$$

Call:

```
lm(formula = miles ~ income + age + kids, data = vacation)
```

Residuals:

Min	1Q	Median	3Q	Max
-1198.14	-295.31	17.98	287.54	1549.41

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-391.548	169.775	-2.306	0.0221 *
income	14.201	1.800	7.889	2.10e-13 ***
age	15.741	3.757	4.189	4.23e-05 ***
kids	-81.826	27.130	-3.016	0.0029 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom

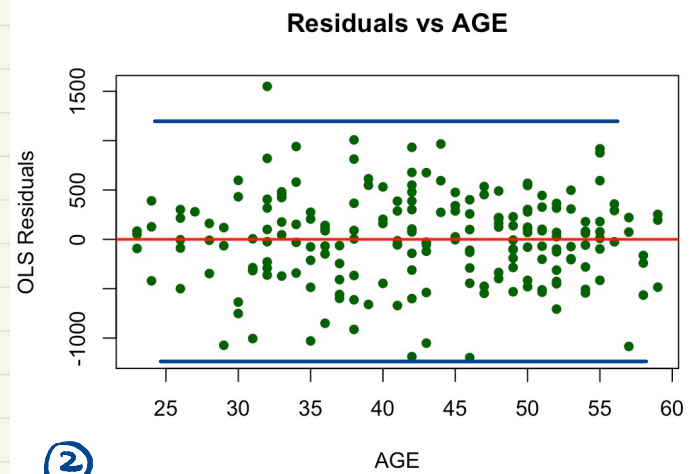
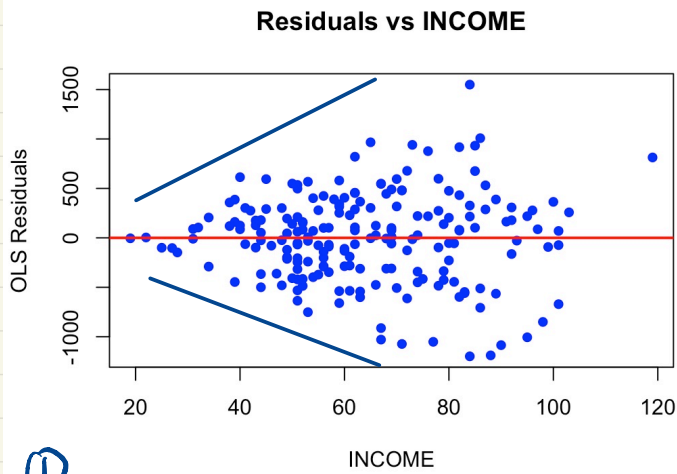
Multiple R-squared: 0.3406, Adjusted R-squared: 0.3305

F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16

	2.5 %	97.5 %
kids	-135.3298	-28.32302

95% CI 不包含 0, kids 的影响在统计上显著为负。

(b)



①: 可以看出圖形呈現喇叭狀, 殘差隨著 income 增加而變大, 存在異質變異性。

②: 看不出特別的圖形, 殘差的分布較均勻, 不存在異質變異性。

(c)

① $H_0: \sigma_{high}^2 = \sigma_{low}^2$ v.s $H_1: \sigma_{high}^2 > \sigma_{low}^2$

② $F = \frac{\sigma_{high}^2}{\sigma_{low}^2} \sim F_{N_{high}-k_{high}, N_{low}-k_{low}}$

③ $\alpha = 0.05$

$RR = \{ F > F_{0.95}(86, 86) = 1.4286 \}$

④ $F^* = \frac{27160654/86}{8750040/86} = 3.1041$

```
> cat("SSE Low (First 90):", sse_low, "\n")
SSE Low (First 90): 8750040
> cat("SSE High (Last 90):", sse_high, "\n")
SSE High (Last 90): 27160654
> cat("G-Q F-statistic:", gq_f_stat, "\n")
G-Q F-statistic: 3.104061
> cat("p-value:", gq_p_value, "\n")
p-value: 1.64001e-07
```

⑤ 因為 $F^* = 3.1041 > 1.4286$, 且 $p\text{-value} < 0.05$, 所以拒絕 H_0 , 有證據支持 $\sigma_{high}^2 > \sigma_{low}^2$, 存在異質變異性。

(d)

t test of coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -391.5480    142.6548  -2.7447 0.0066190 **
income       14.2013      1.9389   7.3246 6.083e-12 ***
age          15.7409      3.9657   3.9692 0.0001011 ***
kids        -81.8264     29.1544  -2.8067 0.0055112 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

95% CI for KIDS (Robust OLS): [-139.323 , -24.32986]

改用 robust standard errors 後, kids 的 standard error 增加,
95% CI 變寬, 但 95% CI 仍不包含 0, 因此與 a 的結論相同。

(e)

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/(income^2))
```

Weighted Residuals:

```
      Min       1Q   Median       3Q      Max
-15.1907  -4.9555   0.2488   4.3832  18.5462
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -424.996    121.444  -3.500 0.000577 ***
income       13.947      1.481    9.420 < 2e-16 ***
age          16.717      3.025    5.527 1.03e-07 ***
kids        -76.806     21.848   -3.515 0.000545 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 6.765 on 196 degrees of freedom
Multiple R-squared:  0.4573,    Adjusted R-squared:  0.449
F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

95% CI for KIDS (Conventional GLS):

```
> print(ci_gls_conv)
      2.5 %    97.5 %
kids -119.8945 -33.71808
```

95% CI for KIDS (Robust GLS): [-121.4134 , -32.19919]

Conventional GLS 和 Robust GLS 的 95% CI 皆比 (a) & (d) 的 OLS
95% CI 還要窄, 表示給予變異致正確的权重後, 能夠提高估計的
效率

Conventional GLS 和 Robust GLS 的 95% CI 的差異不大, 表示假設
 $\sigma^2 = \sigma^2_{INCOME}$, 可以解決異質變異致的問題。